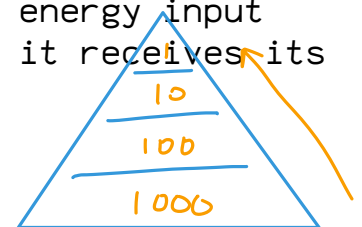


unit 8

- **Kinesis**—non-directional movement in response to a stimulus
- **Taxis**—directional movement in response to a stimulus
- **Chemotaxis**—stimulus is a chemical agent
- **Phototaxis**—stimulus is light
- **Behavior**—an animal's reaction to internal and external stimuli
- **Animal Communication**—how animals communicate to attract mates, defend territory, scare predators, help group members...
 - ▶ **Visual Cues**—a visual cue communicates to others, such as colorful patterns on poison dart frogs, warning predators that they will die if they eat the frog
 - ▶ **Audible Cues**—an audible cue communicates to other animals, such as wolves howling to call to other pack members
 - ▶ **Chemical Cues**—ants release a death pheromone when crushed, chemically signaling to nearby ants to be alert; danger is near
 - ▶ **Tactile Cues**—baboons groom each other and touch to show affection
 - ▶ **Electrical Cues**—electric eels communicate using weak electrical field in murky, unclear water
- **Cooperative Behavior**—increases the fitness of the individual and the population, meaning that cooperative behaviors are evolutionarily “fit”
- **Altruistic Behavior**—behaviors that decrease the fitness of the individual but increase the fitness of the population
- **Food web**—shows all feeding interactions
- **Food chain**—shows just one line of feeding interactions (arrow points to the animal receiving the energy, the consumer)
- **Endotherm**—regulate internal body temperature
- **Ectotherm**—rely on environment for internal body temperature
- **Metabolic Rate per unit Body Mass**—how fast an organism is performing metabolism. Inversely related to the size of an organism (little=high metabolic rate per unit body mass)
- **Growth**—organisms grow when energy input/flow is greater than energy loss
- **Death**—organisms die when energy loss is greater than energy input
- **Trophic Level**—the level of an organism based on when it receives its energy and which organisms it feeds off of
 - ▶ **Producer**—creates their own food
 - ▶ **Primary Consumer**—eats producers
 - ▶ **Secondary Consumer**—eats primary consumers
 - ▶ **Tertiary Consumer**—eats secondary consumers
 - ▶ **Apex Consumer**—does not have a predator
 - ▶ **10% Rule**—only 10% of an organism consumed's energy carries over into the next trophic level, and every trophic level is only around 10% of the biomass of the level below it
- **Predator**—animal that eats another for food
- **Prey**—animal that is eaten by another for food
- **Autotroph**—creates their own food
- **Photoautotroph**—creates their own food via light (photosynthesis)
- **Photosynthesis**—creating food from light
- **Chemosynthesis**—creating organic compounds from inorganic chemical compounds



- **Heterotroph**—consumers other organisms for food
- **Herbivore**—heterotroph that just eats plants
- **Omnivore**—heterotroph that eats both plants and other animals
- **Carnivore**—heterotroph that only eats animals
- **Detritivore**—physically ingests dead matter and breaks it down physically
- **Decomposer**—chemically breaks down dead matter
- **Scavenger**—feeds off of dead animals

***Look at math pdf for Population Growth Equations

- **Limiting Factor**—limits a population's size
- **Density-Dependent Limiting Factor**—dependent on the density of the population (ex: disease, competition)
- **Density-Independent Limiting Factor**—density has no effect on the size of the population (flood, natural disaster)
- **Carrying Capacity (K)**—the amount of individuals an environment can sustain
- **Logistic Growth**—limited resources and limiting factors keep a population to a carrying capacity
- **Exponential Growth**—population growth does not slow down due to no limiting factors and unlimited resources
- **Community**—all biotic factors in an area
- **Biotic factor**—living thing
- **Abiotic factor**—no living thing
- **Ecosystem**—all biotic and abiotic factors in an area
- **Biome**—regions with similar ecosystem makeup around the globe
- **Simpson's Diversity Index**—calculates how biodiverse an ecosystem is through species richness and species evenness
- **Biodiversity**—variety of life found in a particular area; genetic diversity, ecosystem diversity, and species diversity
- **Species Richness**—how many species there are in an ecosystem
- **Species Evenness**—how evenly species are distributed throughout the population

$$D_s = 1 - \sum \left(\frac{n}{N} \right)^2$$

Simpson's Diversity Index (b/w 1-0)

$$D_s = 1 - \sum \left(\frac{n}{N} \right)^2$$

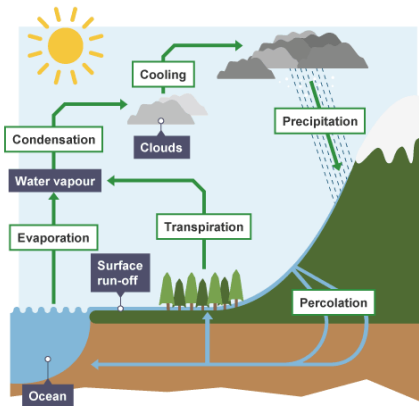
0—low biodiversity (one species)
1—high biodiversity

$1 - \left(\left(\frac{3}{7} \right)^2 + \left(\frac{3}{7} \right)^2 + \left(\frac{1}{7} \right)^2 \right)$
 0.61

- **Trophic Cascade Model/Food Pyramid**—shows feeding habits and biomass and energy levels of each trophic level
- **Resource Partitioning Model**—taking different niches allows for minimized competition so multiple similar species can survive
- **Symbiosis**—at least one animal benefits, a relationship between two animals
 - **Mutualism**—both animals benefit
 - **Commensalism**—one animal benefits, the other is not affected
 - **Parasitism**—one animal benefits off the harm of another
- **Diversity and Resilience**—when there is greater genetic, species, and

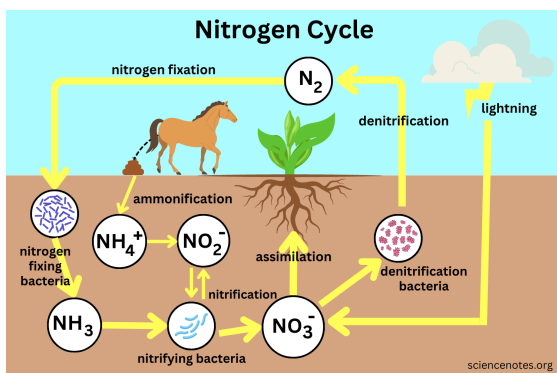
ecosystem diversity, there is a better chance that there is a favorable adaptation that is “fit” for a changing environment, thus making populations more resilient to change

- **Keystone Species**—ecosystem relies on a species’ presence (wolves in Yellowstone). Typically predators. Without keystone species, another population would explode (that which the keystone species consumes), overuse resources, and the environment would suffer, reducing biodiversity. Keystone species keep biodiversity.
- **Invasive Species**—no natural predators and unlimited resources when introduced to a foreign environment. Exhibit exponential growth. Harmful to the environment, reducing room for other species.
- **Geological/Meteorological Impact**—extinctions as a result of random events can reduce biodiversity
- **Biogeochemical Cycles**—explain how different vital elements cycle throughout the environment
- **Oxygen Cycle**—plants use CO_2 and release O_2 . Animals take released O_2 and release CO_2 .
- **Water Cycle**—



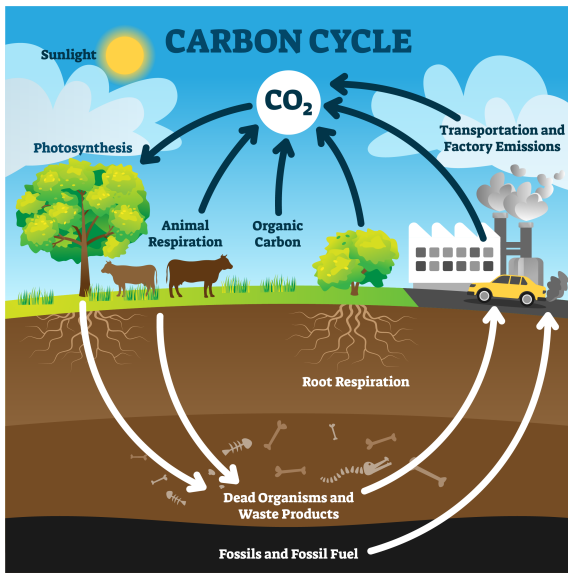
- **Transpiration**—water evaporation from plants through stomata
- **Evaporation**—water turning to a gas and rising to cloud level
- **Condensation**—water condensing into clouds
- **Precipitation**—water falling back to earth from clouds

- **Nitrogen Cycle**—



- **Nitrogen Fixing**—turns N_2 into ammonia (NH_3).
- **Nitrification**—turning ammonia or nitrite into nitrate
- **Assimilation**—nitrate is uptaken by plants
- **Denitrification**—turning nitrate into N_2

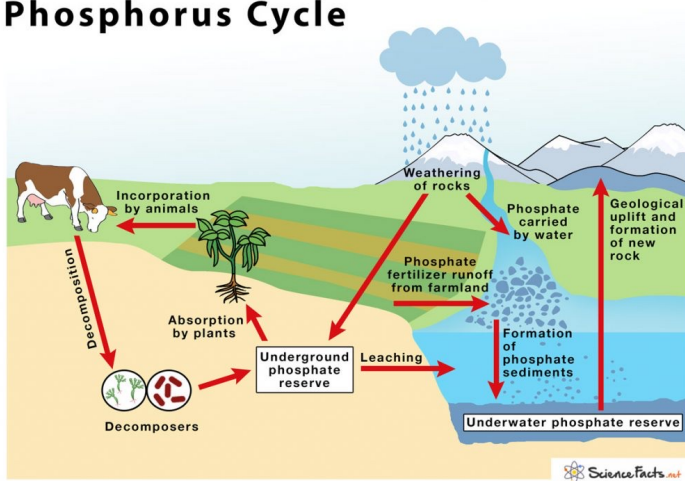
• Carbon Cycle–



- ▶ **Greenhouse Gas Emissions**—mass artificial emissions of CO_2 that imbalances the natural carbon cycle (too much CO_2 in the atmosphere) from burning fossil fuels
- ▶ **Fossil Fuels/Dead Organisms**—rich in carbon

• Phosphorus Cycle–

Phosphorus Cycle



*** ATMOSPHERE IS NOT INVOLVED

- ▶ **Weathering**—rocks slowly weather, or erode, due to weather. This releases phosphorus.
- ▶ **Runoff**—fertilizers rich in phosphorous end up in water
- ▶ **Algal bloom**—algae proliferates and kills all other marine life once too much phosphorus runs off into a water source